Distributed Edge Artificial Intelligence

A Thesis (Major Project) submitted in partial fulfilment of the requirements

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

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2026

Declaration

We, the students of Computer Science & Engineering, Barak Valley Engineering College, declare that the project report entitled **"Distributed Edge Artificial Intelligence"** has been prepared under the guidance of **Mr. Rana Singha**, Computer Science & Engineering, Barak Valley Engineering College.

The report has been prepared without resorting to plagiarism. We have adhered to the principles of academic honesty and integrity. No falsified or fabricated data have been presented in this report. Further, the matter embodied in this project report has not been submitted previously for the award of any degree or diploma.

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Abstract

Distributed Edge Artificial Intelligence (Edge AI) places artificial intelligence workloads across devices near data sources, edge servers, and cloud resources. This report presents a preliminary literature study of how such workloads can be distributed while accounting for communication, computing capacity, energy use, and privacy. A thematic review is used to examine end-edge-cloud architectures, federated learning, collaborative inference, communication efficiency, resource heterogeneity, and security considerations. The reviewed work indicates that distributing computation closer to data can support responsive applications and reduce dependence on centralised processing, but the benefits depend on task placement, network conditions, and device capabilities. Federated training and partitioned inference are identified as complementary approaches, each with trade-offs in communication, coordination, and resource requirements. The project remains at the literature-review and feasibility-assessment stage; no implementation or experimental measurements are claimed. Future work will select an application and platform, then evaluate an appropriate distributed-AI design using reproducible performance and energy measures.

Keywords: Distributed Edge AI, Edge Intelligence, Federated Learning, Collaborative Inference, End-Edge-Cloud Computing

Acknowledgement

We would like to extend our sincere thanks to all those who helped us prepare this report. Without their guidance, support, cooperation, and encouragement, we would not have been able to complete it successfully.

We express our sincere gratitude to our project guide, **Mr. Rana Singha**, for valuable guidance, support, and encouragement throughout this work.

We also thank the faculty members of the Department of Computer Science & Engineering, Barak Valley Engineering College, for their suggestions and academic support.

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Chapter 1

Introduction

Artificial intelligence (AI) is increasingly used in applications that process data generated by mobile devices, sensors, and Internet of Things (IoT) systems. Sending all such data to a distant cloud for analysis can require sustained network communication and may not suit applications with limited bandwidth or strict response-time requirements. These pressures have encouraged interest in performing at least part of the computation closer to where the data is produced.

Distributed Edge Artificial Intelligence studies how AI workloads can be placed across resource-constrained end devices, nearby edge servers, and cloud infrastructure. Rather than treating every device as an independent system, distributed approaches allow multiple nodes to contribute to training, inference, or other application tasks. The resulting design must decide which tasks to execute at each layer, what information to exchange, and how to coordinate nodes with different capabilities.

1.1 Background

Edge AI refers to executing artificial intelligence computations close to the data source, at the network edge, rather than in a centralised cloud data centre [2]. Edge resources may include end-user devices, gateways, and nearby edge servers, working alongside centralised cloud resources. These layers differ in proximity to users, compute capacity, storage, and network connectivity. A system may therefore allocate lightweight or latency-sensitive work to nearby devices and use more powerful resources for tasks that exceed their local capacity.

Edge intelligence encompasses both the use of AI to manage edge systems and the execution of AI workloads on edge resources [2]. Distributed Edge AI extends this setting by coordinating work across interconnected, heterogeneous nodes. For example, a distributed scheduler can place application tasks across edge nodes as system conditions change [1]. In collaborative inference, a model's computation can be partitioned so that multiple devices contribute to producing a result [3]. These approaches make task placement and coordination important parts of the system design.

The choice of where to execute a workload affects more than processing speed. Sending data or intermediate model outputs across a network consumes bandwidth and introduces communication delay; keeping computation local can reduce some transfers but is constrained by device capacity and energy availability. Federated learning offers a way for clients to train collaboratively without collecting their raw training data at one central

location, while still requiring the exchange and coordination of model updates [2]. Partitioned inference, in turn, must balance local computation against the cost of transferring intermediate results [3].

There is no single placement strategy that fits every workload. A design that is appropriate for a delay-sensitive application may not be optimal when energy, network availability, or device diversity is the main constraint. Distributed systems must account for communication cost, heterogeneous computing resources, changing node availability, and the privacy and security implications of sharing data or model information [2, 1]. This report reviews these considerations as a basis for selecting a suitable application and implementation direction in a later project phase.

1.2 Problem Statement

The central problem considered in this project is how to distribute AI training and inference across end devices, edge resources, and cloud infrastructure when those resources have different capabilities and communication links have a cost. Centralised execution can require data to travel away from its source, whereas fully local execution may exceed the computing or energy capacity of an individual device. Distributing work therefore introduces trade-offs among response time, communication, resource use, and coordination.

This report examines those trade-offs through a preliminary literature review and feasibility assessment. It focuses on end-edge-cloud architectures, federated training, collaborative inference, communication efficiency, resource heterogeneity, and privacy and security considerations. The specific application and implementation platform have not yet been selected, and no system has been implemented or experimentally evaluated as part of this stage. The review is intended to identify relevant design choices and provide a grounded basis for defining the next phase of work.

1.3 Objectives

The objectives of this preliminary study are:

1. Review distributed edge-AI architectures and approaches to federated training and collaborative inference.

2. Identify key challenges related to communication, heterogeneity, privacy, energy, and available hardware resources.

3. Establish a feasibility-assessment framework to guide selection of an application and platform for future implementation and evaluation.

1.4 Scope and Report Organization

The scope of this report is a literature-stage study of distributed edge AI; it does not claim a completed implementation, selected application, or new experimental measurements. The review considers architecture, training, inference, communication, heterogeneity, privacy and security, and hardware and energy constraints. Chapter 2 surveys the related literature. Chapter 3 describes the review protocol and proposed feasibility-assessment and technical approaches. Chapter 4 presents the literature synthesis and discusses feasibility and limitations. Chapter 5 summarises the conclusions and identifies future work.

The project follows a six-phase research and development process: (1) learning the fundamentals of Edge AI and Distributed Edge AI, (2) reviewing research papers and building a verified reference collection, (3) conducting small-scale experiments, (4) assessing the feasibility of a larger project, (5) implementing the selected system, and (6) testing and refining the implementation. At the current stage, the work focuses on the first two phases. The subsequent phases will be undertaken as understanding of the field develops and a suitable application is identified.

Chapter 2

Literature Survey

2.1 From Cloud AI to Edge Intelligence

The reviewed literature positions distributed edge AI as a response to the limits of centralised cloud processing. Edge AI performs AI computation near the user at the network edge rather than in a centralised data centre; this shift has been enabled by improvements in AI efficiency, the proliferation of IoT devices, and the growth of edge computing. Zhou et al. describe edge intelligence as extending AI to the "last mile" for mobile and IoT devices that generate data at the edge. Deng et al. distinguish AI for edge from AI on edge, with the latter covering model training and inference on edge devices [2]. The term "edge" has no universally accepted definition; its relation to cloud computing depends on proximity, capacity, and device heterogeneity [2].

2.2 End-Edge-Cloud Architectures

A common architectural pattern distributes processing across end devices, edge servers, and the cloud. End devices handle lightweight local computation, while edge servers provide nearby, more capable resources. The literature distinguishes the complementary roles of AI for edge and AI on edge, and describes orchestration of heterogeneous on-device, edge, and cloud resources for distributed AI. Collaborative learning across cloud and edge resources is also studied [2]. Hu et al. describe a distributed edge-AI scheduler that places application tasks across heterogeneous nodes while accounting for changing system conditions [1].

2.3 Distributed Training: Federated Learning

Federated learning (FL) trains local models on multiple clients without collecting raw user data at a central data centre. Federated learning and knowledge distillation are described as training approaches, while model splitting and partitioning support inference. FL challenges include statistical heterogeneity, communication cost, resource allocation, and privacy and security. Communication can dominate computation when local datasets are small and participant processors are comparatively fast [2].

Communication efficiency, resource allocation, and data heterogeneity are recurring concerns in distributed edge learning [2].

2.4 Distributed Inference: Partitioning and Collaboration

Model-partition methods split a deep neural network between a device and server based on transmission latency and device energy. Early-exit methods can reduce communication when intermediate features are sufficient. DistrEdge applies deep reinforcement learning to distribute CNN inference across heterogeneous IoT edge devices and reports a 1.1–3× speedup over the compared methods [3]. Edge-intelligence surveys also cover the complementary roles of distributed training and inference [2].

2.5 Communication Efficiency

Communication is a central bottleneck across training and inference. What is transmitted affects per-round overhead and the number of communication rounds. Quantisation and pruning, together with in-network computation, are discussed as ways to address communication cost. Computation-intensive inference may require task splitting across edge devices and servers, making communication efficiency especially important [2, 3].

2.6 Heterogeneity, Non-IID Data, and Stragglers

FL over edge nodes must address device heterogeneity, non-independent and identically distributed (non-IID) data, and limited communication resources. Synchronous FL can wait for the slowest worker, while asynchronous FL may cause frequent model transfers. The straggler effect can limit synchronous training time. Non-IID and imbalanced data can cause local updates to drift, resulting in instability and slower convergence; system heterogeneity can also lead to stragglers and dropouts. Participant selection is one proposed mitigation, while edge-assisted architectures, data augmentation, and mediator-based rescheduling are reported as approaches to non-IID bias. These challenges are part of the broader edge-intelligence research landscape [2].

2.7 Privacy and Security

FL does not eliminate privacy risks: a malicious participant may infer sensitive attributes from shared models, and systems may be vulnerable to data or model poisoning. One example in the reviewed literature reports inferring whether an individual's input appeared in a gender-classifier training set with up to 90% accuracy. Other surveyed threats include honest-but-curious servers, free-riding participants, poisoning, and model inversion. Inference attacks may recover training samples; differential privacy is one mitigation, with

a possible trade-off against model performance. Surveys classify poisoning and inference attacks and identify communication blockage, poisoning, and backdoor attacks among high-priority threats. Privacy-preserving cloud-edge-end architectures include encrypted or anonymised protocols and multi-centre hierarchical FL. Blockchain-based FL is also discussed as a way to remove a central aggregator.

2.8 Energy and Hardware Constraints

Latency, energy efficiency, and computation or memory footprint are important evaluation metrics; latency-sensitive applications may have constraints around 100 ms. Energy-efficient edge training is another motivation in the literature; one cited example uses AlphaGo's reported requirement of 280 GPUs and a $3,000 electricity cost per game to illustrate the cost of computation. Reported Raspberry Pi 5 measurements for MobileNetV3-Small include 2.14–2.17 ms latency, approximately 2.55 W net power, and 5.5 mJ per inference. Reported ResNet-18 measurements include 12.9–14.1 ms, 2.4–2.8 W, and 31–39 mJ per inference.

Other reported results include up to 10 W for a Jetson Nano GPU configuration, approximately 5 W for a Google Coral TPU, and approximately 2 W for an Intel NCS2 accelerator. Reported power values for an Arduino Nano, Raspberry Pi, and idle Jetson Nano are 0.17 W, 3.5 W, and 1.3–1.6 W, respectively. Another comparison reports a Jetson Nano approximately three times faster and 1.67 times lower in energy than a Raspberry Pi for its network configuration. These results use different platforms, workloads, and protocols and are indicative rather than directly comparable.

2.9 Application Domains

Reviewed applications include autonomous and connected vehicles, smart homes, industrial automation, healthcare, surveillance, real-time gaming, smart factories, and precision-agriculture pest detection.

Chapter 3

Methodology

Because the project remains at the learning and literature-review stage and the application is undecided, the methodology below is proposed rather than executed.

3.1 Review Protocol

The preliminary study follows a structured literature-review approach. Research questions are modelled on existing reviews of computing, storage, and communication technologies in distributed edge AI and optimisation along the edge-fog-cloud data path. The approach also draws on systematic reviews that develop taxonomies of edge AI systems and examine cloud-edge AI integration [2, 1]. Sources are selected from academic databases for relevance to architecture, training, inference, communication, heterogeneity, privacy and security, energy, and applications.

3.2 Analytical Framework

Sources are synthesised across eight themes: definition and motivation; architecture; training; inference; communication; heterogeneity; privacy and security; and energy and hardware. This structure is intended to help map a later application choice to the relevant literature.

3.2.1 Six-Phase Research and Development Plan

The project is organized into six sequential but iterative phases.

Phase 1: Learning About Edge AI. Develop an understanding of Edge AI, Distributed Edge AI, end-edge-cloud architectures, distributed inference, federated learning, communication constraints, and resource management.

Phase 2: Literature Review and Reference Collection. Study relevant research papers, compare existing methods, record verified bibliographic details, and identify potential research gaps and application opportunities.

Phase 3: Small-Scale Experiments. Investigate selected technical concepts through limited experiments, such as local inference, distributed execution, or communication-overhead measurements, depending on the findings of the initial research.

Phase 4: Feasibility Assessment. Compare candidate applications according to technical relevance, implementation complexity, available resources, cost, and measurable

evaluation criteria.

Phase 5: Project Implementation. Design and build a prototype for the application selected by the team following the preceding research and feasibility assessment.

Phase 6: Testing and Refinement. Evaluate the prototype against appropriate functional and performance requirements, analyze limitations, and refine the implementation.

The project is currently in Phases 1 and 2. The specific application, implementation platform, and final architecture have not yet been selected. Phases 3–6 therefore represent planned future work rather than completed activities.

3.3 Proposed Feasibility-Assessment Framework

A multi-criteria assessment is proposed using latency, energy efficiency, and computation and memory footprint, extended with bandwidth sensitivity, scalability, and cost. Candidate platforms could be compared using a reproducible benchmarking approach that measures latency, throughput, power, and energy.

3.4 Proposed Technical Methodology

Two possible implementation directions are proposed for a later phase:

1. **Partitioned collaborative inference:** split a model by layer or horizontally across devices and exchange intermediate outputs, following published partitioning schemes [3].

2. **Federated training:** participants train local models and share model updates through an aggregation server, as surveyed for mobile edge networks and edge training [2].

Neither direction has been implemented; both remain proposals for subsequent work.

The approaches described above are preliminary candidates identified from the literature, not finalized implementation choices. Their suitability will be investigated through further literature analysis, small-scale experimentation, and feasibility assessment. The final approach will be selected based on the identified problem, available resources, implementation requirements, and evaluation objectives.

Chapter 4

Results and Discussion

As this is a preliminary, literature-stage study, the results are the outcomes of the review and feasibility synthesis rather than experimental findings.

4.1 Literature Synthesis

The reviewed literature converges on layered end-edge-cloud architectures, with federated training and partitioned or collaborative inference as two major technical families [2, 3]. Communication efficiency is a recurring bottleneck across both.

4.2 Feasibility Indications

The literature suggests that distributed edge AI can support latency- sensitive and privacy-sensitive workloads, while remaining constrained by communication and energy. In one reported setup, edge-server inference increased from 0.123 seconds at 1 Mbps to 2.317 seconds at 50 kbps, while a mobile device took more than 2 seconds. Hardware energy figures vary substantially by device and workload.

Table 4.1: Hardware metrics reported in the reviewed literature.

Platform or workload	Metric	Reported value
Raspberry Pi 5, MobileNetV3-Small	Latency, net power, energy	2.14–2.17 ms; about 2.55 W; 5.5 mJ per inference
Raspberry Pi 5, ResNet-18	Latency, net power, energy	12.9–14.1 ms; 2.4–2.8 W; 31–39 mJ per inference
NVIDIA Jetson Nano	Peak power	Up to 10 W
Google Coral TPU	Power	About 5 W
Intel NCS2 and Raspberry Pi 3B+	Accelerator and host power	About 2 W and 3 W
Arduino Nano	Power	0.17 W
Raspberry Pi	Power	3.5 W
NVIDIA Jetson Nano	Idle power	1.3–1.6 W

Partitioned collaborative inference is reported to achieve 1.1–3× speedups over the methods compared in DistrEdge [3]. A communication- efficient FL method reports up to 87% improvement in convergence time and up to 3.6% improvement in accuracy relative

to standard FL. These results come from different hardware, workloads, and measurement protocols and should not be compared directly.

4.3 Discussion and Limitations

The synthesis indicates a trade-off between latency and energy on one side and communication cost on the other. Moving computation closer to the data may reduce latency and bandwidth pressure, but lower edge computing capacity and device heterogeneity can require model partitioning, compression, or collaborative execution. Non-IID data and slow or unreliable nodes can create stragglers and dropouts. Privacy and security risks, including inference and poisoning attacks, also constrain how data and model updates are handled.

No implementation or measurements have been carried out; this feasibility assessment relies entirely on published results. Those results use different platforms, workloads, and measurement setups. Even power-measurement protocols vary; one study reports reducing measurement uncertainty by factors of 6 and 12 on two device families relative to an established protocol. Direct cross-study comparison of energy values must therefore be treated cautiously. The field is still developing, and the definition of "edge" remains unsettled. Open challenges include resource constraints, security threats, scalability, and communication overhead.

Chapter 5

Conclusion and Future Work

This preliminary report has reviewed distributed edge AI and assessed feasibility at a conceptual level. The literature describes a shift from centralised cloud AI toward layered end-edge-cloud architectures, with federated learning and collaborative DNN partitioning as major technical families. Potential suitability for latency- and privacy-sensitive applications is bounded by communication cost, device and data heterogeneity, energy constraints, and privacy and security risks [2, 3].

The project is currently in the foundational learning and literature-review stages. The immediate work involves strengthening the understanding of Distributed Edge AI, analyzing relevant research, and organizing a verified reference collection. The next steps will involve small-scale experiments, followed by a feasibility assessment to select a suitable application and implementation platform. After the project direction has been finalized, a prototype will be developed, tested, and refined. No implementation or experimental results are claimed at the present stage.

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The links above lead to the online publication records. Local copies are stored in the `references/` folder.